CSE276 · ARTIFICIAL INTELLIGENCE FOUNDATIONS

UNIT 1

Foundations of AI and AI Problem Solving

What intelligence means for a machine, where the field came from - and how you turn a messy real-world problem into something a computer can actually search.

History & Evolution • Characteristics & Types • AI vs ML vs DL vs DS • Intelligent Systems & Lifecycle • Applications • Problem Formulation • State Space • Performance Measures

What This Unit Covers - Two Halves

Part A is the landscape. Part B is the machinery - and it is what Unit II runs on.

- PART A - Foundations of Artificial Intelligence
  - Introduction to AI: its history and evolution
  - Characteristics and types of Artificial Intelligence
  - AI vs Machine Learning vs Deep Learning vs Data Science
  - Intelligent systems and the AI lifecycle
  - Applications: healthcare, agriculture, finance, education, manufacturing, robotics, smart cities
- PART B - Artificial Intelligence Problem Solving
  - AI problem formulation and state-space representation
  - Problem characteristics
  - Search space and solution space
  - Performance measures
  - Overview of AI-based problem-solving approaches

01

SECTION

Introduction to AI: History and Evolution

IN THIS SECTION

- What counts as 'intelligent' - and why the answer keeps moving
- The four schools: thinking / acting, humanly / rationally
- The Turing Test and its limits
- Seven eras from McCulloch-Pitts to Generative AI
- Why AI Winters happened

Which of These Is 'Artificial Intelligence'?

UNIT 1

Same question, four answers - and the honest answer changes every decade.

A

A thermostat

- Reads temperature, switches the heater
- Follows exactly one rule
- Nobody calls this AI today

B

Google Maps route

- Searches millions of possible routes
- Re-plans live around traffic
- You call it 'the app', not 'the AI'

C

Face unlock

- Learns your face from examples
- Works in the dark, with a beard, with glasses
- Was research-grade AI in 2014

D

ChatGPT / Gemini

- Writes essays, code and proofs
- Everyone calls this AI
- Ask again in 2035...

SHOW OF HANDS Vote A / B / C / D. Then defend your line: what exactly disqualifies the ones you rejected?

Defining AI: The Rational Agent View

KEY CONCEPT · The definition your exam actually wants

Artificial Intelligence the field of building systems that perceive their environment and take actions that maximise the chance of achieving their goals.

Four classical definitions AI has been defined along two axes - THINKING vs ACTING, and the HUMAN standard vs the RATIONAL standard.

Thinking humanly cognitive modelling - replicate the actual human reasoning steps.

Thinking rationally the 'laws of thought' - formal logic and provable inference, from Aristotle onwards.

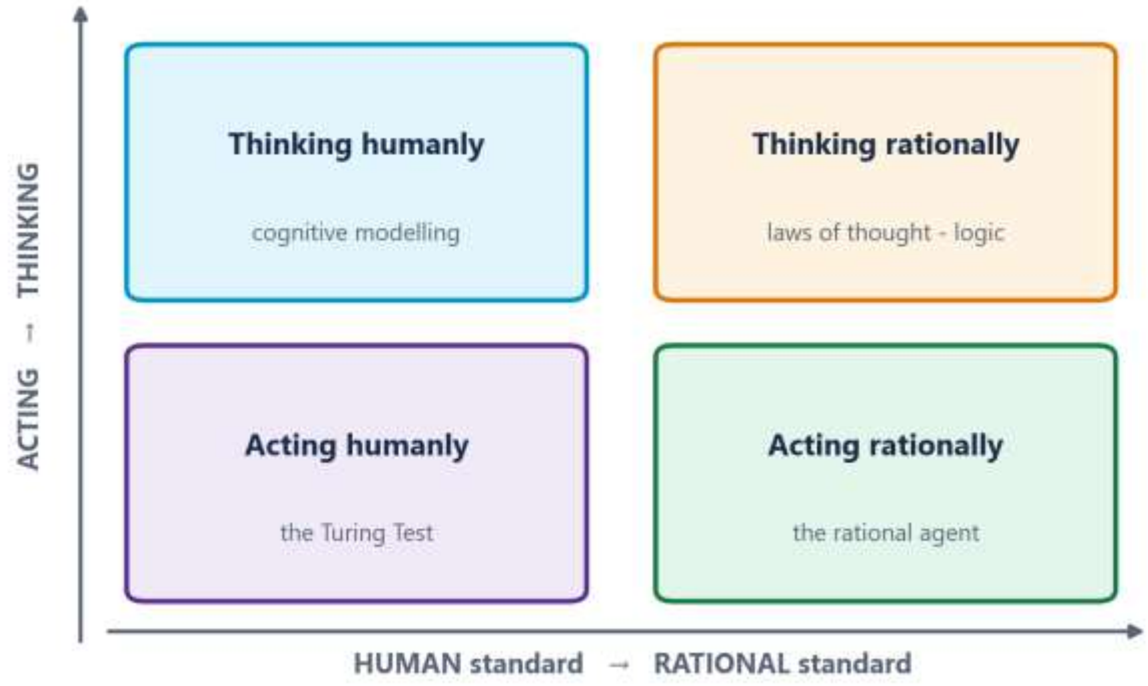
Acting humanly the Turing Test - be indistinguishable from a person in conversation.

Acting rationally the RATIONAL AGENT view - do the right thing, by whatever mechanism. This is where modern AI, and this course, lives.

Why 'acting rationally' won

- It is MEASURABLE - you can score whether the right action was taken
- 'Acting humanly' imports human biases and human mistakes as the target
- It does not require us to first solve psychology or consciousness

The Four Schools, on One Diagram



HOW TO READ THIS FIGURE

Two axes thinking vs acting; the human standard vs the rational standard.

Four boxes every historical definition of AI sits in one of them.

Bottom right modern AI, and this course.

Why it won 'the right action' is measurable; 'like a human' imports human error as the target.

TAKEAWAY **AI is defined by ACTING RATIONALLY - the bottom-right box.**

Characteristics of Artificial Intelligence

UNIT 1

Six properties that, together, distinguish an AI system from ordinary software.

CHARACTERISTIC 1

Perception

- Takes in information about its environment
- Cameras, microphones, sensors, keystrokes, logs
- Without perception there is no agent - only a calculator

CHARACTERISTIC 2

Reasoning & inference

- Draws conclusions that were not explicitly stored
- Rule chaining, logical deduction, probabilistic inference
- The output was not in the input

CHARACTERISTIC 3

Learning & adaptation

- Improves with experience rather than staying fixed
- From data, or from feedback and consequences
- Not required for ALL AI - expert systems do not learn

CHARACTERISTIC 4

Goal-directed behaviour

- Acts to achieve an objective, not just to respond
- Has a performance measure it is trying to maximise
- This is what makes it an AGENT rather than a function

CHARACTERISTIC 5

Handling uncertainty & incompleteness

- Works with noisy, partial, contradictory information
- Probability, heuristics, approximation
- Real problems are never fully specified

CHARACTERISTIC 6

Autonomy

- Operates without step-by-step human instruction
- Degrees of autonomy, not a yes/no property
- More autonomy demands more trust - and more testing

The Turing Test - Brilliant, and Not Enough

How the Imitation Game works

Setup a human interrogator holds a text conversation with two hidden parties - one human, one machine.

Pass condition the interrogator cannot reliably tell which is which.

Turing's move (1950) stop asking 'can machines think?' - it is undefinable. Ask a question you can actually run.

Capabilities implied natural language, knowledge representation, automated reasoning, and machine learning.

Total Turing Test adds vision and robotics, so the machine must also perceive and manipulate objects.

Why modern AI moved past it

It rewards deception the winning strategy includes faking typing errors and dodging hard questions.

Not constructive passing tells you nothing about HOW to build a better system - no engineering signal.

ELIZA (1966) a 200-line pattern-matching script convinced users it understood them. Fooling people is easier than thinking.

Human is a strange target it is a low bar in some ways and an impossible one in others.

Modern replacement task benchmarks - ImageNet, GLUE, MMLU - which are measurable and comparable.

Evolution of AI - Part 1: Birth and the First Winter

AI's history is not a smooth curve - it is boom, bust, boom, bust, boom.

1943-52

Gestational period

McCulloch & Pitts model the first artificial neuron

Hebbian learning (1949)

Turing's paper and the Turing Test (1950)

1956

AI is named

Dartmouth Workshop - McCarthy, Minsky, Rochester, Shannon

The term 'Artificial Intelligence' is coined

A 2-month, 10-man study... that ran 60 years

1956-74

Golden era

LISP (McCarthy, 1958)

General Problem Solver (Newell & Simon)

ELIZA, SHRDLU, and the first search algorithms

Wild over-promising to funders

1974-80

First AI Winter

Lighthill Report (UK, 1973) is damning

DARPA cuts funding

Cause: COMBINATORIAL EXPLOSION - no compute for the search

Evolution of AI - Part 2: Expert Systems to Generative AI

UNIT 1

1980-87

Expert systems boom

- MYCIN diagnoses blood infections
- XCON saves DEC around $40M/year
- Knowledge, not search: IF-THEN rules from experts
- A billion-dollar industry

1987-93

Second AI Winter

- LISP machine market collapses - cheap PCs won
- Expert systems were brittle and could not learn
- The knowledge acquisition bottleneck

1993-2011

Quiet, statistical AI

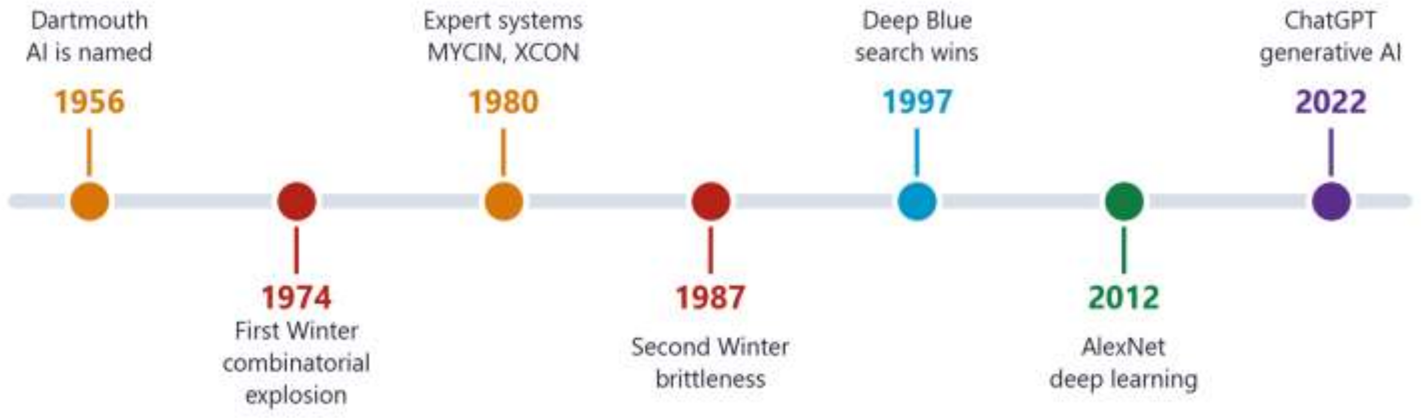
- Deep Blue beats Kasparov (1997)
- Focus shifts to probability and machine learning
- The internet finally supplies data

2012-now

Deep learning & GenAI

- AlexNet crushes ImageNet using GPUs (2012)
- Transformers - 'Attention Is All You Need' (2017)
- AlphaGo (2016), AlphaFold (2020)
- ChatGPT, Gemini, Stable Diffusion

Seventy Years in One Line



HOW TO READ THIS FIGURE

The shape, not the dates two booms, two winters, and the boom we are in.

Winter 1 (1974) COMBINATORIAL EXPLOSION - good algorithms, impossible compute.

Winter 2 (1987) BRITTLENESS and the knowledge acquisition bottleneck.

1997 Deep Blue - the peak of search plus encoded knowledge. Your Unit II.

2012 onward data + GPUs + better algorithms, together.

TAKEAWAY **Winter 1 is exactly why heuristic search exists - and why Unit II is about searching SMARTLY.**

Why Did AI Winters Happen? (And Could One Happen Again?)

COMMON PITFALL · **The pattern repeats: over-promise, under-deliver, defund**

The mechanism researchers over-promise to secure funding, hardware and data cannot deliver, funders lose confidence, money vanishes, progress stalls for a decade.

Winter 1 (1974-80) COMBINATORIAL EXPLOSION. The algorithms were sound but required compute that did not exist.

Winter 2 (1987-93) BRITTLENESS and the KNOWLEDGE ACQUISITION BOTTLENECK - hand-coded rules do not generalise and do not scale.

What is different now AI is commercially profitable at scale. The winters occurred when funding was grant-dependent; revenue is a sturdier foundation.

What could still trigger a correction unmet AGI promises, an energy or compute cost wall, a high-profile safety failure, or regulation following public harm.

THINK-PAIR-SHARE 60 seconds with your neighbour: name one AI promise being made publicly today that you think will NOT be delivered by 2030.

Case Study: Deep Blue vs Garry Kasparov, 1997

REAL-WORLD APPLICATION · **The match that made the public believe - and a lesson about search**

What happened IBM's Deep Blue defeated reigning world champion Garry Kasparov 3.5-2.5 - the first time a computer beat a world champion under tournament conditions.

How it won custom hardware evaluating around 200 million board positions per second, ALPHA-BETA PRUNED SEARCH, and an evaluation function hand-tuned by grandmasters.

The search connection Deep Blue is, at heart, a state-space search agent with a very good heuristic. That is exactly what you build in Unit II.

What it was NOT Deep Blue did not learn. No memory of past games, no understanding of chess. Brute-force search plus human expertise encoded by hand.

The 2017 contrast AlphaZero learned chess from nothing but the rules through self-play in four hours, and beat the best hand-crafted engine in the world.

The lesson Deep Blue is the peak of 'search plus encoded knowledge'. AlphaZero is 'learn it from data'. Your Units II and IV, in one match.

CHECK YOUR UNDERSTANDING

Quick Check - Section 1

UNIT 1

QUICK CHECK

A system passes the Turing Test by convincing a human judge it is human. What does this prove about the system?

- **A** It thinks rationally, using formal logic
- **B** It can act in a way indistinguishable from a human in conversation
- **C** It possesses genuine consciousness and understanding
- **D** It has achieved Artificial General Intelligence

ANSWER: B The Turing Test measures ACTING HUMANLY - external behaviour only. ELIZA (1966) fooled users with simple pattern matching and understood nothing. Behaviour is not evidence of understanding, consciousness, or generality.

02

SECTION

Types of
Artificial Intelligence

IN THIS SECTION

- Axis 1 - Capability: ANI, AGI, ASI
- Axis 2 - Functionality: Reactive, Limited Memory, Theory of Mind, Self-Aware
- Why every system you have ever used is Narrow AI
- Where the research frontier actually sits

Classification by Capability - How Broad Is the Intelligence?

Axis 1 asks: how WIDE is the competence?

AXIS 1 · LEVEL 1

Artificial Narrow Intelligence (ANI)

- Also called Weak AI
- Excellent at ONE task; useless outside it
- Cannot transfer learning to a new domain
- Siri, Google Translate, AlphaFold, Maps routing, every chatbot
- STATUS: 100% of AI in existence today

AXIS 1 · LEVEL 2

Artificial General Intelligence (AGI)

- Also called Strong AI
- Human-level ability across ANY domain
- Transfers knowledge between unrelated tasks
- Common-sense reasoning, self-directed learning
- STATUS: hypothetical. No system is close.

AXIS 1 · LEVEL 3

Artificial Super Intelligence (ASI)

- Surpasses the best human minds in every field
- Better at science, art, strategy and social skill
- Raises the alignment and control problem
- STATUS: theoretical. Philosophy, not engineering.

POLL GPT-4 writes code, passes exams, analyses images and translates. Is it still Narrow AI? Justify your answer.

Capability: Three Rungs, and We Are on the Bottom One

every real system you have ever used is on the bottom rung

HOW TO READ THIS FIGURE

Breadth, not depth this axis asks how WIDE the competence is.

ANI excellent at one task, useless outside it. All of it.

AGI transfers knowledge across unrelated domains. Does not exist.

ASI beyond the best humans everywhere. Philosophy.

The trap 'narrow' says nothing about sophistication.

TAKEAWAY AlphaFold beats every human at protein folding and cannot play tic-tac-toe. That gap IS narrowness.

Classification by Functionality - How Deep Is the Model of the World?

Axis 2 asks: how DEEP is the machine's model of the world, and of other minds?

Type	Core characteristic	Learns?	Real-world example
Reactive Machines	No memory of the past. Reacts only to the current input.	No	Deep Blue, spam filters, basic thermostats
Limited Memory	Retains recent history for a short window and uses it to decide.	Yes (trained)	Self-driving cars, chatbots, all modern ML
Theory of Mind	Understands that others have beliefs, emotions and intentions.	Research	Social / companion robots - active research, not solved
Self-Aware	Possesses consciousness and a model of its own internal state.	Theoretical	Does not exist. Philosophy, not engineering.

CLASSIFICATION DRILL

Classify These Four Systems - Both Axes

UNIT 1

THE PROBLEM

- A self-driving car changing lanes on a highway
- IBM Deep Blue playing chess in 1997
- Google Maps computing your route home
- The AI in a science-fiction film that fears being switched off

WHY IT MATTERS

- Exam questions almost always ask for BOTH axes - answer with both
- 'Narrow' describes breadth, not sophistication
- If a system learns from data, it is at least Limited Memory

STEP-BY-STEP SOLUTION

Self-driving car Capability = ANI (it only drives). Functionality = LIMITED MEMORY - it tracks surrounding vehicles over the last few seconds to predict their paths.

Deep Blue Capability = ANI (chess only). Functionality = REACTIVE MACHINE - evaluates the current board with no memory of earlier games.

Google Maps Capability = ANI. Functionality = LIMITED MEMORY - it uses recent traffic history, and learns typical speeds per road per hour.

Sci-fi AI Capability = AGI or ASI. Functionality = SELF-AWARE. And it exists only in fiction - which is exactly why public intuition about AI is so badly calibrated.

ANSWER **Every real system above is ANI. Only the fictional one is not.**

CHECK YOUR UNDERSTANDING

Quick Check - Section 2

UNIT 1

QUICK CHECK

A warehouse robot navigates aisles, remembers the last 30 seconds of obstacle positions, and cannot do anything except move stock. Classify it on BOTH axes.

- A AGI + Limited Memory
- B ANI + Reactive Machine
- C ANI + Limited Memory
- D ANI + Theory of Mind

ANSWER: C ANI because it does exactly one task and cannot transfer. Limited Memory because it retains a short rolling history of obstacles - it is not purely reactive, but it builds no permanent world model and models no other minds.

03

SECTION

AI vs Machine Learning vs Deep Learning vs Data Science

IN THIS SECTION

- The nested hierarchy - and where Data Science overlaps rather than nests
- When learning from data beats writing rules
- Why deep learning needed 2012, not 1986
- The one example that proves AI is bigger than ML

The Nested Hierarchy - Four Terms, One Structure

UNIT 1

AI ⊃ ML ⊃ DL. Data Science overlaps all three without sitting inside them.

OUTERMOST

Artificial Intelligence

- Any technique that makes machines behave intelligently
- Includes rules, logic, SEARCH - not only learning
- A hand-written IF-THEN chess engine is AI with zero ML
- Born 1956

SUBSET OF AI

Machine Learning

- Systems that learn patterns y = f(x) from data
- No explicit rules written by a programmer
- Needs humans to engineer the features
- Your Unit IV

SUBSET OF ML

Deep Learning

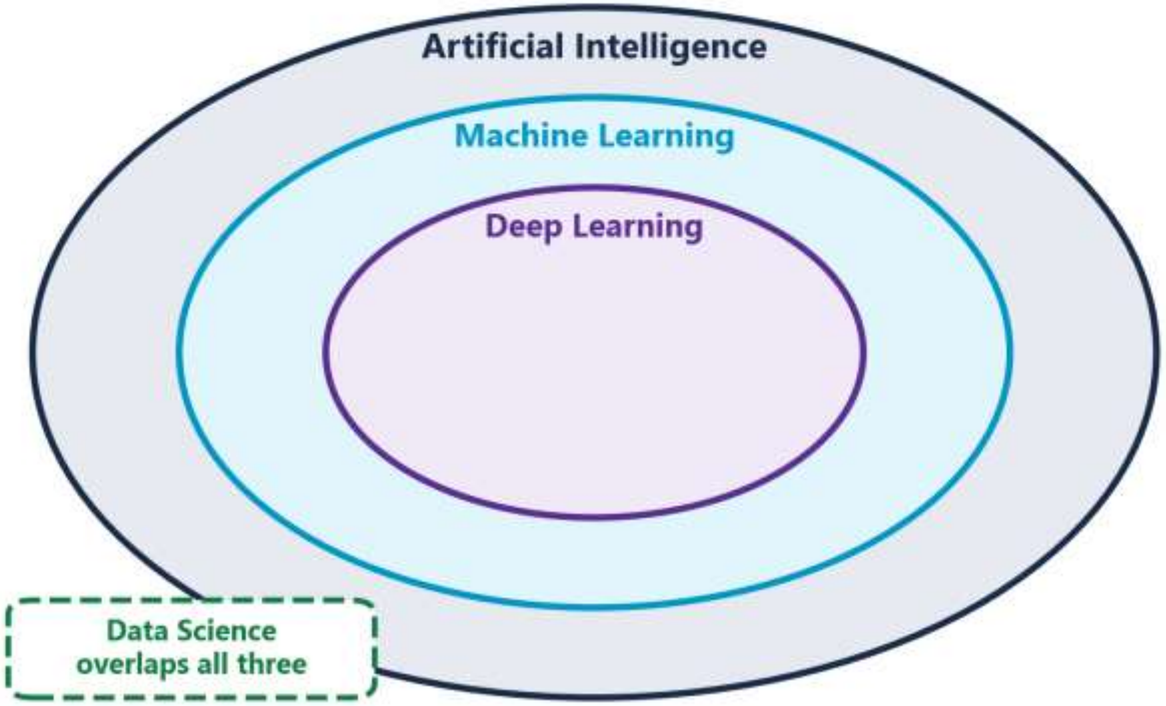
- Neural networks with many hidden layers
- Learns the FEATURES itself from raw pixels / audio / text
- Needs big data and GPUs
- Powers vision, speech and LLMs - Units IV and V

OVERLAPPING FIELD

Data Science

- Extracting insight from data for decisions
- Statistics + domain expertise + engineering + ML
- Includes dashboards and A/B tests that use no AI
- NOT a strict subset - it overlaps AI

The Nested Hierarchy, Drawn



HOW TO READ THIS FIGURE

AI any technique that makes a machine behave intelligently - rules and SEARCH included.

ML ⊂ AI learns patterns from data instead of being given rules.

DL ⊂ ML neural networks that learn the features themselves.

Data Science the dashed box - overlaps rather than nests.

For THIS course Units II and III are almost entirely AI WITHOUT machine learning.

TAKEAWAY **If you think AI means ML, two thirds of this syllabus will not make sense to you.**

Traditional Programming vs Machine Learning

UNIT 1

The direction of the arrow reverses. That is the whole idea of machine learning.

	Traditional Programming	Machine Learning
What YOU write	The rules, explicitly, by hand	Nothing - you supply examples instead
Inputs to the process	Data + Rules	Data + Answers (labels)
What comes out	Answers	Rules (the trained model)
Spam example	IF subject contains 'FREE' AND has 3+ links THEN spam	Show it 100,000 labelled emails; it derives the pattern itself
Fails when	Rules get too numerous or too subtle to write	Data is scarce, biased, or unrepresentative

TRY IT Write the rules to distinguish a cat from a dog in a photo. 60 seconds. Go.

CHECK YOUR UNDERSTANDING

Quick Check - Section 3

UNIT 1

QUICK CHECK

A team builds a hand-coded IF-THEN system that diagnoses equipment faults from sensor thresholds. It contains no learning of any kind. Is it AI?

- A No - without machine learning it cannot be AI
- B Yes - it is AI, specifically a rule-based expert system
- C Only if the thresholds were derived statistically
- D It is Data Science, not AI

ANSWER: B AI is any technique that makes a machine behave intelligently - including hand-written rules. This is a classic expert system, exactly the technology that drove the 1980s AI boom (MYCIN, XCON) - and exactly what you build in Practical 3. ML is a SUBSET of AI, not a synonym.

04

SECTION

Intelligent Systems and the AI Lifecycle

IN THIS SECTION

- What makes a system 'intelligent' rather than merely automated
- The six phases of the AI development lifecycle
- Why it is a loop, not a line
- Where AI projects actually fail

What Makes a System 'Intelligent'?

Both are 'automatic'. Only one of them decides.

Automated system

Behaviour fixed. Same input, same output, forever.

Knowledge entirely in the code the programmer wrote.

Uncertainty cannot cope. Undefined input, undefined behaviour.

Adaptation none. Change requires a developer.

Examples a washing machine cycle, a payroll system, a traffic light on a fixed 90-second timer.

Intelligent system

Behaviour depends on perceived context and on a goal.

Knowledge has an explicit knowledge base or a learned model it can reason over.

Uncertainty degrades gracefully - uses probability, heuristics, defaults.

Adaptation improves with data or feedback, or at minimum re-plans.

Examples adaptive traffic signals that re-time on live density, a diagnostic assistant, a route planner.

DECIDE A lift that goes to the floor you press: automated or intelligent? What if it predicts demand and pre-positions itself?

1.4 AI LIFECYCLE

The Artificial Intelligence Lifecycle - Six Phases

UNIT 1

It is a LOOP, not a line - monitoring in phase 6 sends you back to phase 1.

01 PROBLEM SCOPING

- Define the goal and success criteria
- Decide the task type
- Ask: does this even need AI?

02 DATA / KNOWLEDGE ACQUISITION

- Collect data, OR elicit rules from experts
- Check licensing, privacy and consent
- Is it representative of reality?

03 PREPARATION

- Clean, encode, structure
- Formalise the knowledge base
- Often the largest single time cost

04 MODEL / SYSTEM BUILD

- Choose the approach: search, rules, or learning
- Build a simple baseline first
- Then improve

05 EVALUATION

- Test on UNSEEN cases
- Use the right performance measure
- Check failure modes, not just averages

06 DEPLOY & MONITOR

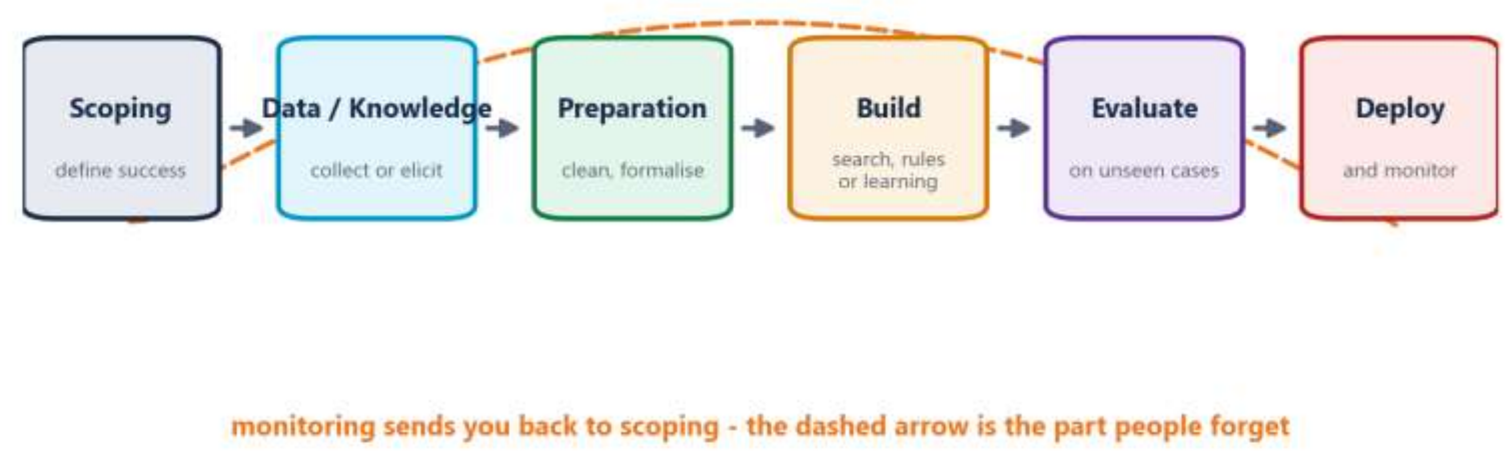
- Integrate, serve, and observe
- Track drift and degradation
- Feed findings back to phase 1

PREDICT Which phase consumes the most time in a real project? Vote before I reveal it.

1.4 AI LIFECYCLE

The AI Lifecycle Is a Loop, Not a Line

UNIT 1



HOW TO READ THIS FIGURE

Phase 2 is dual collect DATA for a learning system, or elicit RULES from an expert.

The bottleneck eliciting rules has its own famous problem - experts cannot always say what they know.

Preparation usually the largest single time cost.

Phase 4 choosing search vs rules vs learning is THE engineering decision.

The dashed arrow deployment is not the end. The world changes.

TAKEAWAY If nobody can say what decision changes when the system works, stop at phase 1.

Where AI Projects Actually Fail

COMMON PITFALL · **Most failures happen before anything is built**

Failure 1 - Solving the wrong problem the system works perfectly and nobody uses it, because no decision depends on its output.

Failure 2 - No success criterion 'improve the student experience' cannot be measured, so the project can never be finished or failed.

Failure 3 - Wrong tool for the problem using machine learning where search or a rule would do - or the reverse. Choosing the approach is an engineering decision, not a fashion decision.

Failure 4 - Unrepresentative knowledge or data the rules came from one expert with one style; the data came from one city.

Failure 5 - No monitoring the system silently degrades for months until somebody notices a business metric has moved.

The uncomfortable truth these are PROCESS failures, not algorithm failures. Better mathematics would not have saved a single one.

05

SECTION

Applications of AI Across Seven Domains

IN THIS SECTION

- Healthcare and agriculture
- Finance and education
- Manufacturing, robotics and smart cities
- What every successful application has in common

1.5 APPLICATIONS

Healthcare and Agriculture

UNIT 1

HEALTHCARE

Diagnostic imaging

- CNNs detect diabetic retinopathy from retinal photographs at specialist-level accuracy
- Deployed where ophthalmologists are scarce
- Also: tumour detection, fracture detection
- Positioned as decision SUPPORT - a clinician signs off

HEALTHCARE

Expert systems & triage

- Rule-based symptom checkers and preliminary triage
- MYCIN's direct descendants
- Transparent reasoning - it can show WHY
- This is your Practical 3

AGRICULTURE

Crop disease detection

- Farmer photographs a leaf; a model names the disease
- Trained on thousands of labelled diseased leaves
- Advice delivered in the local language
- Reaches farmers who will never meet an agronomist

AGRICULTURE

Yield & irrigation

- Yield forecasting from satellite imagery and weather
- Irrigation scheduling from soil moisture sensors
- Pest outbreak prediction
- Water is the scarce resource - scheduling it well matters

Finance and Education

UNIT 1

FINANCE

Fraud detection

- Rule engine + learned model + network analysis, layered
- UPI and card transactions scored in milliseconds
- The hard part is FALSE ALARMS, not detection
- Block a genuine payment and you lose a customer

FINANCE

Credit & risk

- Loan default prediction and credit scoring
- Algorithmic trading and portfolio optimisation
- Heavily regulated - decisions must be EXPLAINABLE
- 'The model said no' is not a legal answer

EDUCATION

Adaptive learning

- Systems that adjust difficulty to the individual student
- Identify which concept a student has actually missed
- Intelligent tutoring systems - one of AI's oldest goals
- Automated grading of structured answers

EDUCATION

Early-warning systems

- Flagging students at risk of failing, so support can reach them
- Built from attendance, submissions, prior grades
- Genuinely useful - and genuinely dangerous
- Who sees the flag, and what happens next?

DISCUSS A model flags you as 'at risk of failing' in week two. Who should be allowed to see that - and what should they do with it?

Manufacturing, Robotics and Smart Cities

Three domains, one shared pattern: sense the world, decide, act before something fails.

MANUFACTURING

Visual inspection

- Computer vision checks PCBs and components for defects
- Thousands of units per hour, never tires
- Detects flaws below the human visual threshold

MANUFACTURING

Predictive maintenance

- Vibration and temperature sensors predict bearing failure
- Replaces fixed-schedule servicing with condition-based
- Unplanned downtime can cost more than $250,000 per hour

ROBOTICS

Autonomous mobile robots

- SLAM - map an unknown space while tracking your position in it
- Warehouse fleets; Amazon runs 750,000+ robots
- PATH PLANNING is a search problem - your Unit II
- Multi-robot coordination - your Unit III

SMART CITIES

Traffic, energy and waste

- Adaptive signals re-time on live vehicle counts
- Electricity demand forecasting and grid balancing
- Leak detection from water pressure anomalies
- Collection routing driven by bin fill sensors

What Every Successful Application Has in Common

KEY CONCEPT · **The pattern behind all seven domains**

1. A narrow, well-defined task detect this defect, predict this failure, plan this route. None of these systems is general - all are ANI.

2. The right technique for the problem search for planning, rules for transparent reasoning, learning for perception. Matching the tool to the problem is the engineering decision.

3. A clear performance measure defects caught, downtime avoided, waiting time reduced. If you cannot measure it, you cannot improve it.

4. Tolerance for imperfection, or a human in the loop the system is allowed to be wrong sometimes - either errors are cheap, or a person reviews them.

5. The cost of being wrong is understood and the system is designed around that cost. Which is a design decision, not a technical one.

APPLY IT Pick any domain from the last three slides. Does your chosen capstone idea satisfy all five? Which one is missing?

CHECK YOUR UNDERSTANDING

Quick Check - Section 5

UNIT 1

QUICK CHECK

A campus wants a system that finds the shortest walking route between any two buildings, accounting for closed paths. Which approach is correct?

- **A** Train a deep neural network on thousands of past walking routes
- **B** Use a large language model and ask it for directions
- **C** Model it as a state-space search problem and run a search algorithm
- **D** Build an expert system with IF-THEN rules for every pair of buildings

ANSWER: C This is a classical SEARCH problem: well-defined states (locations), actions (paths), a goal test and a path cost. BFS or A* solves it exactly, in milliseconds, and is provably correct. A neural network would need data it does not have; an LLM cannot guarantee correctness; and hand-writing rules for every pair does not scale.

06

SECTION

AI Problem Formulation and State-Space Representation

IN THIS SECTION

- The five components of a well-defined problem
- Worked formulation: the vacuum world
- Worked formulation: the 8-puzzle
- State space vs solution space
- Formalising a problem in code

1.6 PROBLEM FORMULATION

Formulating a Problem for AI: The Five Components

UNIT 1

Before you can solve a problem with AI, you must define it formally. These five elements are that definition.

COMPONENT 1

Initial State S_0

- Where the agent starts
- Robot at grid cell (0, 0)
- The 8-puzzle in its scrambled arrangement

COMPONENT 2

Actions A(s)

- The legal moves available in state s
- Note: legal IN THAT STATE, not in general
- {UP, DOWN, LEFT, RIGHT} minus any blocked by a wall

COMPONENT 3

Transition Model Result(s, a)

- Which state results from doing action a in state s
- Result((0,0), RIGHT) = (0,1)
- Together with S_0 and A(s), this DEFINES the state space

COMPONENT 4

Goal Test

- A PREDICATE: is this state a goal? True or false
- Not a state - a function
- Many problems have several goal states

COMPONENT 5

Path Cost c(s, a, s')

- The numeric cost of each individual step
- 1 per move, or metres, or minutes, or fuel
- Sum along a path = the cost of that solution

TOGETHER

= A well-defined problem

- These five are necessary AND sufficient
- A SOLUTION is a path from S_0 to a goal state
- An OPTIMAL solution is the lowest-cost such path

Worked Formulation 1: The Vacuum World

UNIT 1

THE PROBLEM

- Two rooms, A and B
- Each room is either Clean or Dirty
- The vacuum sits in one room at a time
- Actions available: Left, Right, Suck
- Task: formalise this as a state-space problem

WHY IT MATTERS

- The canonical Russell & Norvig example - expect it in the exam
- Note the goal test is a PREDICATE satisfied by two states
- State count 2×2×2 = position × dirtA × dirtB

STEP-BY-STEP SOLUTION

States position × dirt status of both rooms = 2 × 2 × 2 = 8 possible states. Example: [vacuum in A, A dirty, B clean].

Initial state any of the 8 - the agent may start anywhere.

Actions A(s) {Left, Right, Suck} in every state.

Transition model Suck → current room becomes Clean. Left → move to A (no effect if already in A). Right → move to B.

Goal test is every room Clean? Note: TWO of the 8 states satisfy this - vacuum in A with both clean, and vacuum in B with both clean.

Path cost 1 per action, so path cost = number of actions taken.

Optimal solution from [in A, both dirty]: Suck, Right, Suck. Cost 3.

ANSWER **8 states · 2 goal states · cost = number of actions**

1.6 PROBLEM FORMULATION

Worked Formulation 2: The 8-Puzzle

THE PROBLEM

- A 3×3 grid holding tiles 1-8 and one blank
- A tile adjacent to the blank can slide into it
- Start: some scrambled arrangement
- Goal: tiles in order 1-8 with the blank in a fixed position
- Task: formalise it, then size the state space

WHY IT MATTERS

- Modelling the BLANK as the moving piece halves the branching factor
- This is Practical 2 - the 8-puzzle solver using A*
- Note how fast the space grows: 8-puzzle 10^5, 15-puzzle 10^{13}

STEP-BY-STEP SOLUTION

States any arrangement of the 8 tiles plus the blank in 9 cells.

Initial state the given scrambled configuration.

Actions A(s) move the BLANK Up / Down / Left / Right - between 2 and 4 legal moves depending on where the blank sits.

Transition model swap the blank with the adjacent tile in that direction.

Goal test does the arrangement equal the goal arrangement? Here exactly ONE state satisfies it.

Path cost 1 per move, so cost = number of moves.

State space size 9! = 362,880 arrangements - but only HALF are reachable from any given start, so 181,440.

15-puzzle $16!/2 \approx 1.05 \times 10^{13}$ states. Same formulation, hopeless to enumerate.

ANSWER **181,440 reachable states — generated on demand, never listed**

State Space vs Solution Space

Your syllabus names both. They are not the same thing, and exams exploit that.

State space

What it is the set of ALL states reachable from the initial state, together with the transitions between them.

Structure a graph. Nodes are states, edges are actions.

Size usually enormous - 181,440 for the 8-puzzle, ~10^{13} for the 15-puzzle.

Built how implicitly. Generated on demand by applying the transition model - never written out in full.

Defined by S_0 + A(s) + Result(s, a). Those three alone fix the whole space.

Solution space

What it is the set of all PATHS from the initial state to a goal state.

Structure a set of action sequences - each one a candidate answer.

Size can be infinite if the space has cycles and you allow revisits.

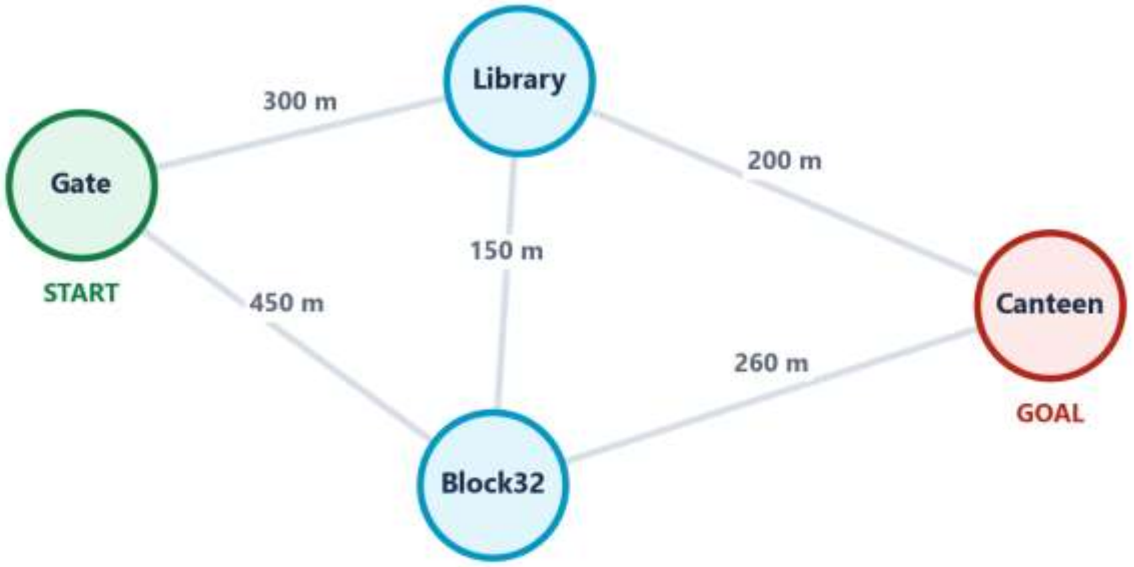
Selected by the goal test decides which paths qualify; the path cost decides which qualifying path is best.

The search task explore as little of the STATE space as possible while still finding a good point in the SOLUTION space.

CHECK For the vacuum world: how large is the state space, and how many distinct solutions exist from [in A, both dirty]?

Campus Routing as a State Space

UNIT 1



HOW TO READ THIS FIGURE

Nodes = states each place you can be standing.

Edges = actions each path you can walk, with its cost in metres.

Initial state Gate. Goal test: are we at Canteen?

A solution any path from Gate to Canteen. An OPTIMAL one is the cheapest.

Try it Gate→Library→Canteen = 500 m. Gate→Block32→Canteen = 710 m. Gate→Library→Block32→Canteen = 710 m.

TAKEAWAY The five components you just learned are exactly this picture - and this picture is Practical 1.

The Same Formulation, in Python

route_problem.py - a state-space problem as a Python class

```
class RoutePlanningProblem:
    def __init__(self, start, goal, road_map):
        self.start = start              # Component 1: initial state
        self.goal = goal                # used by Component 4: goal test
        self.map = road_map             # {place: [(neighbour, distance), ...]}

    def actions(self, state):           # Component 2: legal actions A(s)
        # only the roads that actually leave this place
        return [nbr for nbr, dist in self.map.get(state, [])]

    def result(self, state, action):    # Component 3: transition model
        return action                   # the action IS the place we move to

    def goal_test(self, state):         # Component 4: a PREDICATE
        return state == self.goal

    def step_cost(self, state, action, next_state):
        for nbr, dist in self.map[state]:      # Component 5: path cost
            if nbr == next_state:
                return dist
        return float('inf')

campus = {                          # the state space, defined implicitly
    'Gate':    [('Library', 300), ('Block32', 450)],
    'Library': [('Gate', 300), ('Canteen', 200), ('Block32', 150)],
    'Block32': [('Gate', 450), ('Library', 150), ('Canteen', 260)],
    'Canteen': [('Library', 200), ('Block32', 260)],
}
problem = RoutePlanningProblem('Gate', 'Canteen', campus)
```

WHAT THIS CODE DOES

actions() returns only the roads leaving THIS place - legality depends on the state.

result() a pure function: same state and action always give the same next state. So this environment is DETERMINISTIC.

goal_test() a predicate returning True/False - not a stored state.

step_cost() returns real distances, so path cost is metres. Change it to minutes and the optimal route changes.

- Notice: no search algorithm. Formulation and solution are separate jobs.

07

SECTION

Problem Characteristics

IN THIS SECTION

- Seven questions to ask before choosing a technique
- Decomposability, recoverability, predictability
- Absolute vs relative solutions
- Is the answer a state, or a path?
- The role of knowledge, and of human interaction

The Seven Problem Characteristics

Seven questions. Ask them BEFORE you choose a technique - the answers usually pick it for you.

QUESTION 1

Is the problem decomposable?

- Can it be split into independent smaller problems?
- DECOMPOSABLE: symbolic integration - integrate each term separately
- NOT: the 8-puzzle - moving one tile disturbs the others
- Decomposable problems allow divide-and-conquer

QUESTION 2

Can solution steps be undone?

- IGNORABLE: theorem proving - a useless lemma costs nothing
- RECOVERABLE: the 8-puzzle - you can slide a tile back
- IRRECOVERABLE: chess - a played move is gone forever
- Decides whether you need backtracking or careful planning

QUESTION 3

Is the universe predictable?

- CERTAIN: the 8-puzzle - the outcome of a move is known
- UNCERTAIN: bridge, robot navigation on loose gravel
- Certain outcomes allow full planning in advance
- Uncertain outcomes force plan-act-observe-replan

QUESTION 4

Is a good solution absolute or relative?

- ANY-PATH: answering a factual question - one good answer suffices
- BEST-PATH: the travelling salesman - you need the optimum
- Best-path problems are far more expensive to solve
- Ask whether 'good enough' is genuinely good enough

QUESTION 5

Is the solution a state or a path?

- A STATE: the 8-queens problem - only the final layout matters
- A PATH: route planning - the journey IS the answer
- Changes what you must store during the search
- Path problems need you to remember how you got there

QUESTIONS 6 & 7

Knowledge, and human interaction

- ROLE OF KNOWLEDGE: does the problem need vast knowledge, or just a good search?
- Chess: modest knowledge, huge search
- Newspaper comprehension: enormous knowledge
- INTERACTION: solitary answer, or a conversation with a human?

Applying the Characteristics to Three Problems

Same six questions, three problems, three different profiles - and therefore three different techniques.

Characteristic	8-Puzzle	Chess	Campus route planning
Decomposable?	No - tiles interact	No	Partly - via waypoints
Steps undoable?	Recoverable	Irrecoverable	Recoverable (in planning)
Universe predictable?	Certain	Uncertain (opponent)	Mostly certain
Solution absolute or relative?	Best-path (fewest moves)	Best-path	Best-path (shortest)
State or path?	Path - the move sequence	Path	Path
Role of knowledge?	Low - search dominates	Moderate - evaluation function	Low - the map is the knowledge

YOUR TURN Run the seven questions on ONE of your practicals: the medical expert system. What profile does it have?

CHECK YOUR UNDERSTANDING

Quick Check - Section 7

UNIT 1

QUICK CHECK

In the 8-queens problem you must place 8 queens on a chessboard so none attack each other. Only the final arrangement is reported. Which characteristic does this illustrate?

- **A** The solution is a PATH, not a state
- **B** The solution is a STATE, not a path
- **C** The problem is irrecoverable
- **D** The universe is unpredictable

ANSWER: B Nobody cares in which order you placed the queens - only the final arrangement is the answer. So the solution is a STATE. Contrast route planning, where the sequence of moves IS the answer. This changes what the search must store: state-solution problems need not remember the path.

08

SECTION

Performance Measures and AI Problem-Solving Approaches

IN THIS SECTION

- The four measures: completeness, optimality, time, space
- Branching factor and depth - why complexity explodes
- A worked comparison
- Overview of AI-based problem-solving approaches

Four Performance Measures for Problem-Solving

Four questions you ask of ANY problem-solving method - and the four you will be examined on in Unit II.

MEASURE 1

Completeness

- If a solution EXISTS, is the algorithm guaranteed to find it?
- A complete algorithm never misses an existing solution
- DFS on an infinite branch is INCOMPLETE - it can wander forever
- BFS is complete when the branching factor is finite

MEASURE 2

Optimality

- When it finds a solution, is it the LOWEST-COST one?
- Complete and optimal are DIFFERENT properties
- DFS may find a solution - just not the best one
- Optimality usually costs more time and memory

MEASURE 3

Time complexity

- How many states are generated before a solution is found?
- Expressed with b = branching factor, d = depth of solution
- BFS: O(b^d) - exponential in the solution depth
- This is the combinatorial explosion, quantified

MEASURE 4

Space complexity

- How many states must be held in memory at once?
- Frequently the REAL limit, before time ever becomes one
- BFS: O(b^d) - it stores an entire frontier
- DFS: O(b·d) - only one branch. Far cheaper.

Worked Example: Sizing a Search Before You Run It

UNIT 1

THE PROBLEM

- A delivery robot navigates a campus road network
- Average 4 roads leave each junction, so branching factor b = 4
- The destination is 8 junctions away, so solution depth d = 8
- Question: is exhaustive breadth-first search feasible?

WHY IT MATTERS

- Always estimate b and d BEFORE choosing an algorithm
- Space usually fails before time does
- This calculation is why heuristic search exists - Unit II

STEP-BY-STEP SOLUTION

Step 1 - time BFS generates roughly $b^d = 4^8$ states.

Step 2 - compute it $4^8 = 65{,}536$ states. Entirely manageable.

Step 3 - space BFS also STORES about that many. At ~100 bytes per state, roughly 6.5 MB. Fine.

Step 4 - now change one number Suppose the destination is 15 junctions away instead of 8.

Step 5 - recompute $4^{15} \approx 1.07 \times 10^9$ states. Over a billion.

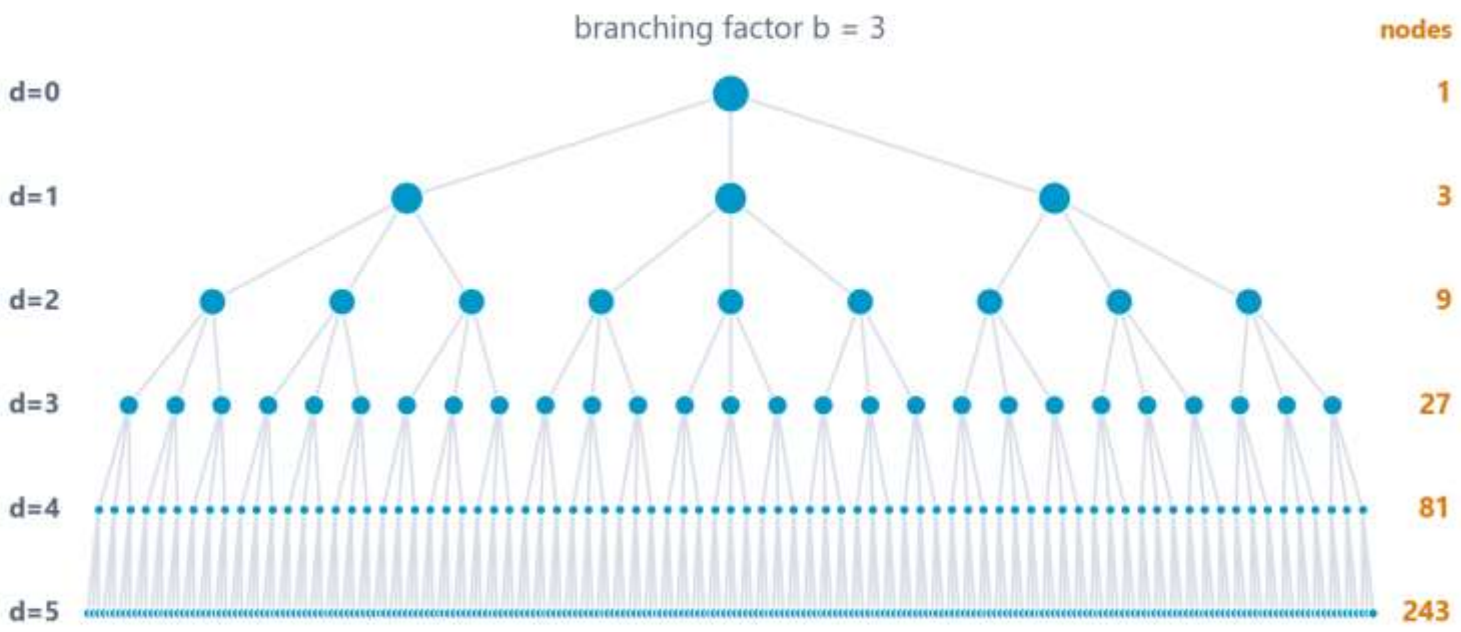
Step 6 - the space bill At 100 bytes each, that is roughly 107 GB of memory - just to hold the frontier.

Conclusion Same algorithm, same campus, seven extra junctions of depth - and it moved from trivial to impossible.

ANSWER $b = 4, d = 8 \rightarrow 65{,}536$ states (fine). $d = 15 \rightarrow \sim 10^9$ states (impossible).

1.8 PERFORMANCE MEASURES

Why b^d Is the Number That Matters



HOW TO READ THIS FIGURE

Each level multiplies the number of nodes by b.

Depth 5 with b = 3 that is already 243 nodes on the frontier.

Depth 10 59,049. Depth 20: about 3.5 billion.

BFS time AND space both O(b^d) - it must STORE the whole frontier.

DFS space only O(b·d) - one branch. Cheap memory, no guarantee of optimality.

TAKEAWAY One more level of depth multiplies your work by b. That is what 'exponential' actually feels like.

Overview of AI-Based Problem-Solving Approaches

UNIT 1

Four families. Your syllabus is organised around them - and choosing between them is the engineering decision.

APPROACH 1

Search-based

- Explore a state space for a path to the goal
- Uninformed: BFS, DFS. Informed: Best-First, A*, Hill Climbing
- BEST WHEN: states and actions are well defined
- Your Unit II

APPROACH 2

Knowledge-based

- Encode facts and rules; let an inference engine reason
- Semantic networks, frames, production systems, logic
- BEST WHEN: expertise exists and reasoning must be explainable
- Your Unit II second half

APPROACH 3

Agent-based

- Situate the solver in an environment with sensors and actuators
- Reflex, model-based, goal-based, utility-based, learning agents
- BEST WHEN: the system must act continuously in a changing world
- Your Unit III

APPROACH 4

Learning-based

- Infer the rules from data instead of writing them
- Supervised, unsupervised, reinforcement learning; deep learning
- BEST WHEN: humans cannot articulate the rules
- Your Units IV and V

MATCH THEM Route planner · Loan-approval reasoning you must justify · Warehouse robot · Recognising a face. Which approach for each?

1.6-1.8 CONSOLIDATION

Activity: Formulate and Analyse a Problem

UNIT 1

GROUP WORK · 12 minutes

In groups of 3-4, take one problem and produce a complete formal analysis. You are not solving it - you are proving you can DEFINE and DIAGNOSE it.

1

Pick a problem

- Campus emergency evacuation routing
- A robot tidying a room into labelled bins
- Timetabling classes without clashes
- Or propose your own

2

Formulate it

- Initial state, Actions A(s), Transition model
- Goal test - remember, a PREDICATE
- Path cost - and say what units it is in

3

Characterise it

- Run the seven questions
- Decomposable? Recoverable? Predictable?
- State-solution or path-solution?

4

Size and choose

- Estimate b and d - is exhaustive search feasible?
- Which of the four approach families fits?
- Justify your choice in one sentence

DELIVERABLE One group nominates a speaker: 60 seconds giving the five components, one characteristic that surprised you, and your chosen approach with its justification.

UNIT I RECAP

Unit I - The Eight Ideas Worth Remembering

UNIT 1

1 **AI is defined by rational action, not human imitation**
Four classical schools; modern AI acts rationally, because that is measurable and does not import human error as a target.

2 **AI has boomed and busted twice, for identifiable technical reasons**
Winter 1: combinatorial explosion. Winter 2: brittleness and the knowledge acquisition bottleneck. Data, compute and algorithms ended them.

3 **Six characteristics, and two independent classification axes**
Perception, reasoning, learning, goal-direction, uncertainty, autonomy. Capability (ANI/AGI/ASI) and Functionality (Reactive → Self-Aware).

4 **AI ⊃ ML ⊃ DL, with Data Science overlapping**
Not all AI is ML - two thirds of this syllabus is AI without learning. Deep Blue proves it.

5 **An intelligent system decides; an automated one obeys**
And the AI lifecycle is a six-phase LOOP. Most projects fail in phase one, not phase four.

6 **A well-defined problem has exactly five components**
Initial state, Actions A(s), Transition model, Goal test (a PREDICATE), Path cost. Formulation is a design decision - the 8-puzzle proves it.

7 **State space ≠ solution space**
State space is everywhere you could BE, built implicitly. Solution space is every way you could GET THERE. Search minimises the first to find a good point in the second.

8 **Judge any method on four measures**
Completeness, optimality, time $O(b^d)$, space $O(b^d)$. Space usually fails first - and that is precisely why heuristic search exists.

NEXT UP Unit II - Problem Solving using Search Techniques: BFS, DFS, Best First Search, Hill Climbing and A*; then Knowledge Representation, expert systems, logic and Bayesian reasoning.